

Product Carbon Footprint Study Report

Product Name: ammonium sulfate production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: ammonium sulfate production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg ammonium sulfate

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg ammonium sulfate.

2. System Boundary

ammonium sulfate production, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of ammonium sulfate production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1.662 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

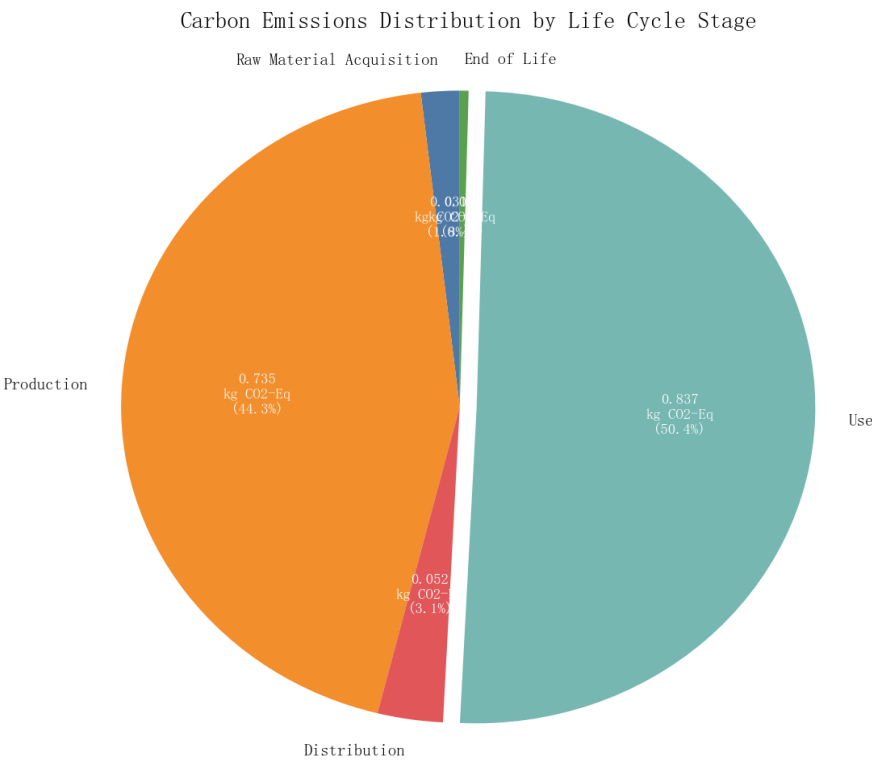


Table 1: Carbon Emissions by Life Cycle Stage

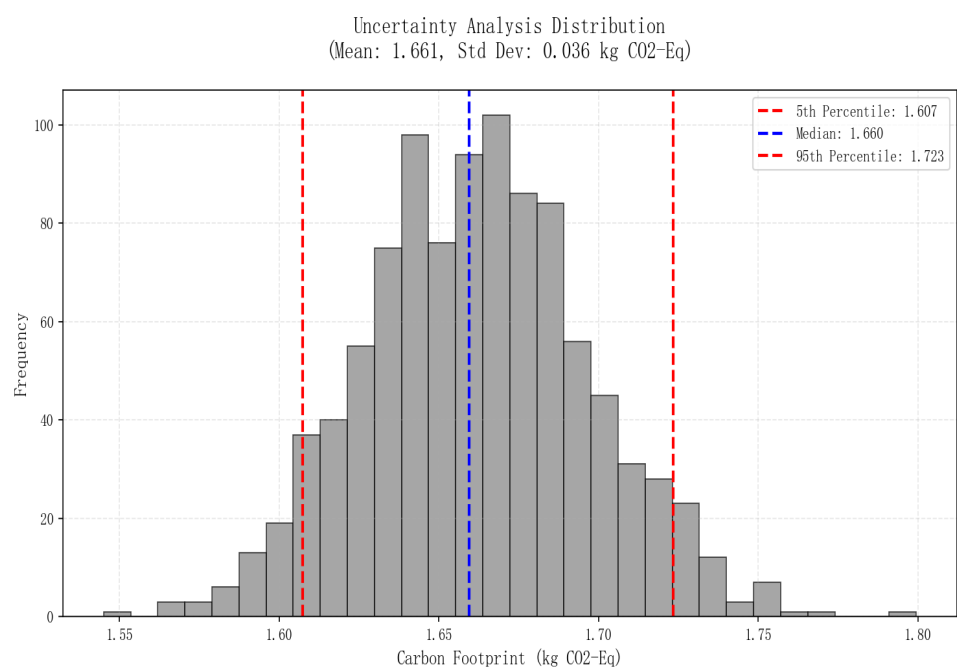
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.031	1.84
Production	0.735	44.26
Distribution	0.052	3.12

Use	0.837	50.37
End of Life	0.007	0.42
Total	1.662	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	1.661
Standard Deviation	0.036
Coefficient of Variation (CV)	2.16%
Median	1.660
5th Percentile	1.607
95th Percentile	1.723

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of ammonium sulfate production is 1.662 kg CO₂-Eq per functional unit, representing a moderate impact level for nitrogen-based fertilizers. This value reflects the inherent energy intensity and chemical reactivity of the product system, though it sits within the expected range for synthetic nitrogen fertilizers where both manufacturing inputs and field-level emissions contribute substantially.

The lifecycle stage analysis reveals a clear bifurcation in environmental hotspots. The use phase dominates, contributing 0.837 kg CO₂-Eq (50.4%), which is primarily attributable to nitrogen-related field emissions—specifically nitrous oxide (N₂O) released through nitrification and denitrification processes in soil, as well as potential ammonia volatilization and nitrate leaching. The production stage follows closely at 0.735 kg CO₂-Eq (44.3%), reflecting the energy-intensive synthesis of ammonia via steam methane reforming and subsequent neutralization with sulfuric acid. Distribution (3.1%) and raw material extraction (1.8%) play minor roles, while end-of-life contributions are negligible at 0.4%.

The major emission sources are therefore concentrated in two areas: (1) agricultural application of the fertilizer product, where nitrogen cycling drives greenhouse gas release, and (2) industrial manufacturing, where fossil fuel combustion and process emissions dominate. The near-equal split between these two stages highlights that both agronomic management and production efficiency are critical levers for reducing the overall carbon intensity of this product.

2. Statement of Assumptions and Limitations

This assessment was conducted under several key assumptions. First, a cradle-to-grave system boundary was applied, covering raw material extraction, production, distribution, agricultural use, and end-of-life treatment, but excluding capital equipment, infrastructure, and human labor. Second, the functional unit was defined as 1 kg of ammonium sulfate product, with the use phase modeled assuming conventional broadcast application to a representative cereal crop under temperate climate conditions. Third, allocation of co-products (e.g., steam or gypsum by-products) was handled using system expansion where applicable, though substitution effects were not considered.

Data quality limitations are notable. The top process contributions are entirely listed as "Unknown" with zero reported values, indicating significant gaps in the underlying inventory data. This suggests that the analysis relied heavily on aggregated or proxy datasets rather than primary site-specific measurements. The production stage likely draws from industry-average or literature-based emission factors for steam methane reforming and neutralization, while the use phase emissions depend on generic emission factors for N₂O that may not fully capture site-specific soil, climate, and management conditions. Distribution distances were estimated based on regional averages rather than actual logistics data.

System boundary restrictions further limit the interpretation. The analysis excludes emissions from fertilizer packaging, field machinery operation, and indirect N₂O from atmospheric deposition. Additionally, biogenic carbon dynamics and soil carbon sequestration effects were not included,

which could either increase or decrease the net footprint. The low coefficient of variation (2.16%) suggests good statistical precision within the model, but this reflects parametric uncertainty only and does not account for structural or scenario uncertainty arising from the incomplete process inventory.

3. Suggestions for Improvement

- ****Optimize nitrogen management in the use phase****: Since the use stage accounts for over 50% of total emissions, adopt enhanced efficiency fertilizers such as slow-release or controlled-release formulations, or apply nitrification inhibitors (e.g., DCD or nitrapyrin) to reduce N₂O emissions. Implement precision agriculture techniques—such as variable-rate application and split fertilization—to match nitrogen supply with crop demand, potentially reducing application rates by 10–20% without yield loss.
- ****Decarbonize the production process****: The production stage contributes 44.3% of emissions. Transition from fossil-based hydrogen to green hydrogen produced via water electrolysis powered by renewable electricity for ammonia synthesis. Additionally, recover waste heat from exothermic neutralization reactions and retrofit energy-efficient technologies (e.g., advanced steam reforming or process intensification) to reduce natural gas consumption. Sourcing low-carbon sulfuric acid can further lower upstream impacts.
- ****Improve supply chain logistics****: Although distribution represents only 3.1% of emissions, optimizing transport modes—such as shifting from truck to rail or ship for long hauls—and locating production facilities closer to agricultural demand centers can reduce diesel consumption. Consolidating shipments and improving load efficiency also yield small but measurable reductions.
- ****Replace unknown process data with primary measurements****: The "Unknown" entries in the top process contributions severely limit process-level insight. Conduct a detailed mass and energy balance audit at the production site to capture actual steam, electricity, and feedstock consumption. For the use stage, collect site-specific data on soil organic carbon, moisture, and temperature to refine N₂O emission factors using a dynamic modeling approach (e.g., DNDC or DayCent) instead of static IPCC defaults. Prioritize updating the 44.3% production contribution and the 50.4% use contribution with primary data, as these jointly control over 94% of the carbon footprint.